TIMOTHY REHM, PHD

timothy.d.rehm@nasa.gov
Washington, District of Columbia, USA

EDUCATION

Brown University, Providence, RI *August 2019 - May 2025*
Doctoral Student in Physics Spring 2025
Master of Science in Physics Spring 2021

Cornell University, Ithaca, NY *August 2015 - December 2018*
Bachelor of Arts in Physics – Cum Laude
Astronomy Concentration
Cornell Tradition Fellow

PUBLICATIONS

– "The EXoplanet Climate Infrared TElescope (EXCITE): A balloon-borne mission measuring spectroscopic phase curves of transiting hot Jupiters," T. D. Rehm, L. Bernard, A. Bocchieri, N. Butler, R. C. Challener, J. Hartley, K. Helson, D. P. Kelly, K. Klangboonkrong, N. Lewis, S. Li, M. Line, S. F. Maher, R. McClelland, L. Mugnai, P. C. Nagler, C. B. Netterfield, V. Parmentier, E. Pascale, J. Patience, L. J. Romualdez, P. A. Scowen, G. S. Tucker, and I. Waldmann, *arXiv:2602:04840*, *Submitted to the Review of Scientific Instruments*, (2026).

– "The design and development status of the cryogenic receiver for the EXoplanet Climate Infrared TElescope (EXCITE)," T. Rehm, L. Bernard, A. Bocchieri, N. Butler, Q. Changeat, A. D'Alessandro, B. Edwards, J. Gamaunt, Q. Gong, J. Hartley, K. Helson, L. Jensen, D. P. Kelly, K. Klangboonkrong, A. Kleyheeg, N. Lewis, S. Li, M. Line, S. F. Maher, R. McClelland, L. R. Miko, L. Mugnai, P. C. Nagler, C. B. Netterfield, V. Parmentier, E. Pascale, J. Patience, L. J. Romualdez, S. Sarkar, P. A. Scowen, G. S. Tucker, A. Waczynski, and I. Waldmann, in *Ground-based and Airborne Instrumentation for Astronomy IX* (Vol. 12184, pp. 830-836). SPIE, (2022).

– "The Exoplanet Climate Infrared Telescope (EXCITE): gondola pointing and stabilization qualification," L. J. Romualdez, T. Rehm et al. in *Ground-based and Airborne Telescopes X* (Vol. 13094, pp. 1914-1922). SPIE, (2024).

– "Assembly, integration, and laboratory testing of the EXCITE spectrograph," L. Bernard, T. Rehm, et. al. in *Ground-based and Airborne Instrumentation for Astronomy X* (Vol. 13096). SPIE, (2024).

– "Integration and testing of a cryogenic receiver for the Exoplanet Climate Infrared Telescope (EXCITE)," A. Kleyheeg, T. Rehm et al. in *Ground-based and Airborne Instrumentation for Astronomy X* (Vol. 13096, pp. 1081-1090). SPIE, (2024).

– "The EXoplanet Climate Infrared TElescope (EXCITE)," P. C. Nagler, T. Rehm et al. in *Ground-based and Airborne Instrumentation for Astronomy IX* (Vol. 12184, pp. 285-295). SPIE, (2022).

– "Design and testing of a low-resolution NIR spectrograph for the EXoplanet Climate Infrared TElescope," L. Bernard, T. Rehm et al. in *Ground-based and Airborne Instrumentation for Astronomy IX* (Vol. 12184, pp. 763-772). SPIE, (2022).

RESEARCH EXPERIENCE

NASA Postdoctoral Program Fellow – NASA GSFC 2026-
– Performing research at NASA Goddard Space Flight Center under the supervision of Dr. Peter Nagler.
– Instrument scientist for the EXICTE mission.

Detector Scientist – Global Science & Technology, Inc./NASA GSFC 2025-2026
– Designed cryo-mechanical testbeds for infrared semiconducting and superconducting detectors using CAD/SolidWorks.
– Analyzed detector data to calculate detector responsivity, dark current maps, and hot pixel levels.
– Operated and performed routine maintenance of test equipment, including temperature controllers, integrating spheres, spectrum analyzers, monochromators, black body sources, vacuum pumps, cryocoolers, and associated drive electronics.

EXoplanet Climate Infrared TElescope (EXCITE) – Brown University 2020-2025
Advisor: Prof. Gregory Tucker, Physics Dept, Brown University
– Designed and fabricated the cryogenic receiver for EXCITE, a NASA-funded balloon-borne near-infrared telescope.
– Integrated cryocoolers to maintain a 100 K and 50 K stage at 130,000 feet using the SuperBIT gondola design for the first time.
– Produced 50+ components using CAD/SolidWorks; personally machined critical parts.
– Leader of the data analysis group, developing Python-based MCMC simulations, signal detrending and systematics modeling.
– Modeled radiometric noise levels and generated spectral images measured by the near-infrared detector. Used detector test data to aid in the design of a data reduction pipeline.
– Built a successful passive heat rejection system using a methanol fluid loop, dissipating 300 W in the 2024 engineering flight.
– Built testbeds for thermal vacuum (TVAC) tests on fluid pumps, electrical components, flight computers, and motors.
– Created component level thermal models in Thermal Desktop to validate cryogenic receiver performance.
– Played a key role in project development and coordinated inter-institutional collaboration throughout the entire mission lifetime.

Tianlai 21 cm Radio Dish/Cylinder Array – Brown University 2020-2021
Advisor: Prof. Gregory Tucker, Physics Dept, Brown University
– Rewrote data analysis pipeline `tlpipe` in Python 3. Pipeline features include RFI extraction, noise source calibration, point source calibration, and complex gain corrections.
– Generated 21 cm Hydrogen maps from radio dish array data using an m-mode map-making formalism.
– Implemented a mode amplitude Legendre decomposition on observed visibilities to improve understanding of the RFI landscape.

Arecibo Pisces-Perseus Supercluster Survey – Cornell University 2018-2019
Advisor: Prof. Martha Haynes, Astronomy Dept, Cornell University
– Reduced Arecibo L-band feed array data in Pisces-Perseus Supercluster using Python-written pipeline tasks and RFI mitigation techniques.
– Computed estimates on galactic peculiar velocities using local density perturbation models.
– Wrote a halo mass group assignment algorithm to analyze the positions of over 30,000 galaxies in the ALFALFA catalog and cross-correlated with SDSS & 2MRS.

Cornell Physics Education Research Lab (CPERL) – Cornell University 2017

Advisor: Prof. Natasha Holmes, Physics Dept, Cornell University
– Performed rigorous data analysis for the Physics Lab Inventory of Critical Thinking survey (PLIC) taken by over 200 undergraduate students.
– Piloted and redesigned Cornell's Introductory Physics Lab sequence to reflect modern Physics education research frameworks.

TEACHING EXPERIENCE & OUTREACH

Student Volunteer – Hope High School Spring 2024
Provide in-class mentorship and support for small group problem-solving and hands-on lab work for over 100 high school physics students.

Sheridan Teaching Seminar Program Certification – Brown University Fall 2019

Graduate Teaching Assistant – Brown University
TA for Physics 0220 *Astronomy* Spring 2020
TA for Physics 0270 *Astronomy & Astrophysics* Fall 2019

Undergraduate Teaching Assistant – Cornell University
TA for Physics 2214 *Oscillations, Waves & Quantum Physics* Fall 2018
TA for Physics 2213 *Electromagnetism* Spring 2017
TA for Physics 1112 *Mechanics & Heat* Fall 2016

INVITED TALKS

Rehm, T., *Phase-resolved spectroscopy of hot Jupiters with the EXoplanet Climate Infrared TElescope (EXCITE)* (2023, May). Talk given at the Carnegie Earth & Planets Laboratory 2023 Spring Astronomy Seminar Series in Washington, DC.

Rehm, T., *Observing Exoplanet Atmospheres with Exoplanet Climate Infrared Telescope (EXCITE)* (2022, August). Talk given at the Emerging Researchers in Exoplanet Science (ERES) VII conference at The Pennsylvania State University, State College, PA.

Rehm, T., *Observing Exoplanet Atmospheres with Exoplanet Climate Infrared Telescope (EXCITE)* (2022, January). Talk given for the Skyscrapers Inc. Astronomical Society of RI in Providence, RI.

Rehm, T. *Testing Measures of Environment in the Pisces-Perseus Region* (2018, June). Talk given to the Undergraduate ALFALFA Team's 2018 Workshop at the Green Bank Observatory in Green Bank, West Virginia.

AWARDS/GRANTS/FELLOWSHIPS

Galkin Foundation August 2024
Brown University Department of Physics Galkin Foundation Fellow
Funding for tuition and graduate stipend for the 2024-2025 academic year.

Forrest Award June 2024
Excellent Work Related to Experimental Apparatus

NASA Rhode Island Space Grant Consortium February 2020
Graduate Student Fellowship 2021
Funding for the 2021 spring and summer to pursue mechanical and thermal modeling of the EXCITE cryostat.

TECHNICAL SKILLS

– Trained in mechanical design, CAD modeling, working with vacuum equipment, fluid transport systems, making electrical harnesses, and systems integration.
– Proficiency in Python, LaTeX, TOPCAT, and SolidWorks.
– Familiarity with MATLAB, Mathematica, and Thermal Desktop.
– Machine Shop Certified at the Brown Joint Engineering & Physics Instrument Shop.

LANGUAGE SKILLS

– Native in English
– Conversational in Spanish & German